$$\ell = -\frac{(n-1)\ln S}{2} - \sum_{t=2}^{n} \frac{(x_t - \theta a_{t-1})^2}{2S}$$

$$\frac{\partial \ell}{\partial \theta} = 0 - \frac{1}{2S} \sum_{t=2}^{n} 2(x_t - \theta a_{t-1})(0 - a_{t-1} - \theta \frac{\partial}{\partial \theta} a_{t-1})$$

$$= \frac{1}{S} \sum_{t=2}^{n} (x_t - \theta a_{t-1})(a_{t-1} + \theta \frac{\partial}{\partial \theta} a_{t-1})$$

$$\frac{\partial^2 \ell}{\partial \theta^2} = \frac{1}{S} \sum_{t=2}^{n} \Big[(-a_{t-1} - \theta \frac{\partial}{\partial \theta} a_{t-1})(a_{t-1} + \theta \frac{\partial a_{t-1}}{\partial \theta})$$

$$+ (x_t - \theta a_{t-1})(\frac{\partial a_{t-1}}{\partial \theta} + \frac{\partial a_{t-1}}{\partial \theta} + \theta \frac{\partial^2 a_{t-1}}{\partial \theta^2}) \Big]$$

$$= \frac{1}{S} \sum_{t=2}^{n} -(a_{t-1} + \theta \frac{\partial a_{t-1}}{\partial \theta})^2 + (x_t - \theta a_{t-1})(2 \frac{\partial a_{t-1}}{\partial \theta} + \theta \frac{\partial^2 a_{t-1}}{\partial \theta^2})$$